

This table consists of materials that can be read for assignment/exams.

Course books

1. Designing Large Language Model Applications, Suhas Pai, March 2025 Publisher(s): O'Reilly Media, Inc.
2. Building LLM Powered Applications: Create Intelligent Apps and Agents with Large Language Models, Valentina Alto, Packt Publishing, 2024
3. Hands-On Large Language Models: Language Understanding and Generation, Jay Alammar and Maarten Grootendorst, 2024

Sl. No.	Topic	Textbook /Chapter
1	Introduction to Large Language Models (LLMs) Overview, use-cases, current trends	2, chapter 1,2 3, chapter 1,2
2	LLM Tools & Frameworks LangChain, LlamaIndex, Ollama, Instructor, Outlines	2, chapter 3
3	Open Source LLMs and Ecosystem Overview	2, chapter 3
4	Prompt Engineering - Part 1 Zero-Shot, Few-Shot Prompting	2, chapter 4 3, chapter 6
5	Prompt Engineering - Part 2 Chain-of-Thought, Prompt Chaining, Adversarial Prompting	3, chapter 6,7
6	Embeddings and Representation Learning Introduction, Semantic Search, Similarity	1, chapter 11
7	Fine-Tuning Embedding Models Base Models, Training Data, Loss Functions, Instruction Embeddings	

8	Chunking Strategies & Vector Databases Sliding Window, Metadata-Aware, Layout-Aware, Semantic, Late Chunking	
9	Conversational Applications and Interpreting Embeddings	Lecture slides
10	Retrieval-Augmented Generation (RAG) Query Enhancement, Context Enrichment	1, chapter 12
11	Advanced RAG Architectures & Explainability	1, chapter 12, Lecture slides
12	AI Agents - Part 1 Agent Definitions, Components, Agentic Workflow	1, chapter 10, Lecture slides
13	AI Agents - Part 2 Agent Loop Prompt, Guardrails, Orchestration Software	1, chapter 10, Lecture slides
14	Advanced Concepts Fine-tuning Gen Models, Model Context Protocol (MCP), Building Agents using MCP	Lecture slides